Internship Management System

CESW281 (E)

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Problem Statement

The current internship application process is mostly traditional where students need to visit companies in person or rely on manual communication through the university this causes fewer opportunities for students , so a centralized digital system can improve speed, transparency, and access by connecting all parties in one organized platform.

Idea Description

The system is a centralized platform that connect students, universities, and companies to make the internship process easier. Students can apply to the opportunities, companies can post and review applications, and universities can track student's progress. It reduces delays, increases transparency, and provides equal access for all students, making the internships more efficient and organized for all parties.

Information Gathering

This report summarizes interviews with three categories involved in internships:

- Students / Interns
- University / Internship Coordinators
- Companies / Employers

The goal is to understand their challenges and needs to improve the internship process.

Category	Name	Job Title	Date of interview
Students / Interns	Mohamad Khaled	Landscape Architecture	18/9/2025
Students / Interns	Sabeen Alorabi	Architectural Engineer	19/9/2025
Students / Interns	Abdullah Alharthy	Industrial Engineer	19/9/2025
Students / Interns	Dana Alzahrani	graphic design	16/9/2025
University / Internship Coordinators	Dr.Noura AlShamrani	Training Coordinator/ Lecturer	21/9/2025
Companies / Employers	Ruyan Aljohani	Senior IT Business Solutions	17/9/2025
Companies / Employers	Raghad Aljohani	Director of product development	18/9/2025

General Questions (for all)

1- How important do you think a platform connecting the university, students, and companies would be?

2- What are the main challenges you currently face in the internship process?

3- Do you have any concerns or feedback about using a digital platform for internships?

4- What features do you think are essential for better communication between students and companies?

Students / Interns

1- How do you usually search for internship opportunities?

2- How would you like to track your progress or evaluation during the internship?

3- What kind of information or services would you like the platform to provide?

4- How was your communication with your academic supervisor or the internship coordinator?

5- Would you like a digital certificate generated directly by the platform?

6- Would you like to see previous students' experiences with a company before applying?

7- Would you like to set internship goals and track them on the platform?

8- What are the biggest difficulties you face when searching or applying for internships?

University / Internship Coordinator

1- Do you currently use a system for managing student internships? If yes, how does it work and how do you communicate with companies through it?
2- Do you face difficulties tracking student placements or those without internships?
3- Would you like the platform to allow full application and tracking of internships?
4- What key features or services would you want in an ideal internship platform?
5- Should performance reports (attendance, evaluation, achievements) be delivered regularly or on demand?

Companies / Employers

1- How do you usually receive internship applications from students?
2- Do you currently have an electronic system to manage student internship requests?
3- What are the main challenges in receiving or filtering internship applications?
4- What student information or skills do you need to see before accepting them?
5- Do you face difficulties in tracking the number of interns or managing their requests each semester?
6- How do you currently monitor intern performance, and would a digital tool make it easier?
7- Would having a dedicated platform account to track interns directly be useful?

From the responses, it is clear that :

Analysis of Companies' Perspective

1- **Application process**

- Most companies rely on LinkedIn for posting training opportunities and receiving applications.

2- **Current electronic system**

- There is no dedicated system for training.
- Companies use their existing hiring systems with the job title labeled as training.

3- **Challenges in receiving and evaluating interns**

- A large number of interviews consumes time and effort.
- In some cases, there is no clear training plan, or the plan does not align with the student's major.

4- **Tracking intern performance**

- Usually done through university or company surveys, or via a department supervisor.
- Some companies find this enough, while others prefer a more digital and easier approach.

5- **Opinion on having a centralized platform**

- Some companies believe LinkedIn is sufficient.
- Others find a platform useful, especially if it requires companies to provide a clear training plan.
- Overall, such a platform would benefit students and universities more, but could also ease companies' workload in receiving and screening applications.

Conclusion

1- The platform could reduce the effort companies spend on managing applications and reviewing candidates.
2- Having a mandatory training plan is a key requirement for the success of the platform.

Analysis of Students' Perspective

1- **Importance of a platform connecting universities, students, and companies**

- Students highlighted that such a platform is very important as it centralizes information about training opportunities.
- It also improves and eases the link between universities and companies.

2- **Challenges in applying for training**

- One of the main challenges is the lack of organization, which causes distraction and makes it difficult to keep track of application updates.
- Difficulty in finding suitable companies, as most companies do not explain the contents and objectives of the training.
- This process often leads to wasted time and effort.

3- **Concerns about using a digital platform**

- Most students have no major concerns.
- The platform would be effective for those who are genuinely motivated to learn.

4- **Preferred methods of tracking progress and evaluation**

- Weekly evaluations from both academic and workplace supervisors.
- Showing the daily logs for completed tasks.

5- **Desired data and services on the platform**

- Information about companies, so that trainees can choose the ones that best fit their goals.
- The ability to apply for training directly.
- Access to previous trainees' experiences.
- These features would make the search more personalized and reliable.

6- **Key features for effective student-company communication**

- A smart search tool that allows students to filter and find suitable training providers.
- Allowing students to set specific criteria to filter the available opportunities.
- This helps companies reduce the number of applicants to those who meet the company's conditions.

7- **Main difficulties in seeking training opportunities**

- Lack of time due to ongoing studies.
- Slow responses from organizations.

8- **Communication with supervisors**

- Students currently rely on informal methods like WhatsApp with academic supervisors.
- This shows the need for more formal and integrated communication tools.

9- **Preferred methods of finding training opportunities**

- Students expressed a clear preference for having a unified platform to focus on entirely.
- This reduces wasted time and effort.

10- **Interest in accessing previous students' experiences**

- All students agreed on the importance of having access to reviews and feedback from former trainees.
- This helps them make informed decisions before applying to a company.

11- **Preference for digital training certificates**

- Students strongly supported the idea of automated digital certificates issued through the platform.
- This ensures students' efforts are officially recognized.

Conclusion

- The results reveal that students face consistent challenges in finding or applying for suitable training opportunities.
- A digital platform that centralizes information, ensures transparency, and provides communication tools would address these issues.
- Key features such as advanced search filters, access to past trainee feedback, and digital certification are highly valued.
- Overall, the responses emphasize the need for a structured, reliable, and student-friendly system that reduces the gap between universities, students, and companies.

Analysis of University's Perspective

- **Current communication methods**

- Email and phone calls are mostly relied on, but they cannot be fully depended on because they are not very efficient, and misunderstandings may occur.

2- **Need for a more efficient platform**

- A dedicated platform is needed to ensure equal and fair opportunities for all students.
- It would also increase the range of companies participating in the training.

3- **Difficulties in finding enough opportunities**

- When the number of students is large, the university faces challenges in finding sufficient training opportunities.

4- **Difficulty in matching skills and qualifications**

- It is difficult to secure opportunities that match students' skills and qualifications.
- This affects their chances to gain meaningful experience.

5- **Digitalization of official documents**

- Converting official documents to digital form protects the rights of students, the university, and the company.
- For example, it allows issuing digital thank-you certificates.

6- **Statistics feature**

- Having a statistics feature provides a clear overview of student performance.
- It also helps the university and companies make improvements.

7- **Student status screen**

- A screen that shows each student's status on demand is very useful for students, the university, and the company.
- It helps track progress, makes follow-up easier, and ensures transparency regarding each student's current status throughout the training process.

8- **Overall benefits of a digital platform**

- Using a digital platform is beneficial because it helps students find training more easily.
- It allows the university to follow up on students and track their progress.
- This makes the whole process easier, more organized, and transparent.

Project Purpose

In many universities, training departments face challenges in tracking students during their internship and checking if they follow both company and university requirements. Most of the work is still done manually or by email, which often causes delays, missing documents, and weak supervision. The purpose of this project is to create a digital platform that connects students, universities, and companies in one system to manage the internship process more effectively. The motivation behind this effort is to make the process simple, fast, and organized, so students can focus on learning and universities can easily follow their progress. This issue is important because, if not solved, it will continue to waste time, reduce the quality of student training, and increase the burden on departments. Therefore, a centralized and secure solution is needed to improve the whole experience.

Project Goals

The main goal of this project is to design and develop a digital platform that connects universities, students, and companies for managing internships. The system helps students equally view training opportunities , submit and get approval for training applications, track their progress, and give clear reports to academic supervisors making the internship process simple, clear, and well-organized.

Findings

• There is an existing platform called “Fursy” (فرصي) that has a similar idea to our system. However, our project aims to improve some parts that are not fully covered in the current platforms.

• Companies mostly use LinkedIn for training applications.

• No clear system exists for managing training.

• Students find it hard to get enough or suitable opportunities.

• Communication is not always clear or organized.

• A single digital platform can make the process easier and fair for all.

• Students want clear training details, reviews, and progress tracking.

• Digital forms and certificates help make everything more official.

Solutions

1. Digital Platform Idea: Create a centralized system for internships.

Description: Students can apply, companies can offer internship training programs, and universities can track progress — all in one place.

2. Digital Tools Idea: Use existing apps to manage internships.

Description: Tools like email, online forms, and task managers help organize the process faster without building a new platform.

3. University Internship Team Idea: Assign a team to handle internships.

Description: The team guides students and communicates with companies, but requires extra staff and may be slower than a digital system.

Project Description

The system is a centralized platform that connect students, universities, and companies to make the internship process easier. Students can apply to the opportunities, companies can post and review applications, and universities can track student's progress. It reduces delays, increases transparency, and provides equal access for all students, making the internships more efficient and organized for all parties.

Project Scope

This project focuses on supporting recent graduates and students who are still pursuing their studies within Saudi universities. It aims to connect them with local companies across the Kingdom to facilitate cooperative training and employment opportunities. The system is designed to operate exclusively within Saudi Arabia, covering universities and organizations located in the country. It does not include training programs outside the Kingdom or for non-local students who are not officially enrolled in Saudi universities.

The Feasibility Study

- **Market Feasibility**

The Internship Management System is useful for students, companies, and universities. Many universities still use paper or email to manage internships, which takes time and causes confusion. This system makes the process easier and faster by putting everything in one place. Students can apply online, companies can share internship opportunities, and universities can easily track the progress of their students. There are some similar systems, but most of them don't connect all three sides. Because of that, many universities and companies will be interested in using this system.

The project has strong market potential because there is a clear need for a simpler and digital way to manage internships.

- **Technical Feasibility**

The Internship Management System is technically feasible and can be built using commonly available tools and technologies. The system can be implemented as a web based application, using HTML, CSS, and JavaScript for the frontend, and PHP or Python for the backend. Data about students, companies, and universities can be stored in a MySQL database. The system can be hosted on a standard university server or on free cloud hosting platforms. In addition, many online resources and tutorials are available, which makes it possible to develop, test, and maintain the system smoothly.

The project is technically feasible and can be successfully created and managed using the available technologies and resources.

- **Operational Feasibility**

The proposed internship management platform is operationally feasible because similar systems in other universities have already shown success. Manual processes often lead to delays, duplicated work, and missing information, but digital platforms proved to reduce administrative workload and improve communication between students, supervisors, and companies. This indicates that adopting such a platform is realistic and acceptable.

- **Schedule and Financial Feasibility**

Based on similar projects, the development of an online internship platform can be completed within a reasonable timeframe. Previous systems were successfully built within 4 to 6 months, including planning, design, implementation, testing, and deployment. Therefore, the project timeline is considered practical and achievable. The project is financially feasible because the required technology and resources are available and affordable. Similar systems showed that costs mainly involve development hours, hosting, and basic maintenance.

For our case, if two developers work part-time over 6 months at an average cost of $1,500 per month the estimated cost is around $18,000 with additional hosting and maintenance costs. Compared to the benefits of reducing paperwork ,saving staff time, and improving student experience this investment is reasonable and justified.

Project Risks and Solutions

1. **Data Loss**
   There is a risk that some data could be lost due to system errors or accidental deletion.

   Solution: Data will be backed up periodically and stored securely in the cloud to prevent loss.

2. **System Downtime**
   The website may experience downtime occasionally due to server or network problems.

   Solution: A reliable hosting service will be used, and the system will be monitored to resolve issues immediately.

3. **Security Problems**
   Unauthorized users or hackers may attempt to access sensitive information.

   Solution: Login authentication, strong passwords, and basic data encryption will be implemented to protect the system.

4. **Lack of Experience**
   Limited experience in developing similar systems may cause delays, errors, or operational difficulties.

   Solution: This risk can be reduced by utilizing available resources, following clear development steps, testing the system regularly, and dividing tasks based on each member's skills.

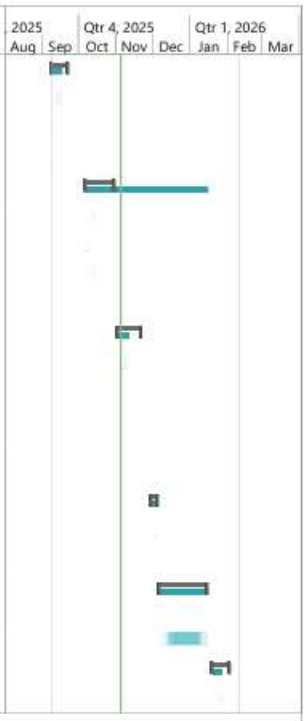
5. **Slow Performance**
   When many users use the system at the same time, performance may decrease.
   Solution: The code will be optimized, large files will be minimized, and reliable hosting will be used to improve system speed.

Executive Summary

The project is feasible and makes sense because it solves real problems in managing internships. Right now, the process is slow, confusing, and relies too much on manual work. By creating a digital platform, we can connect students, companies, and universities in one place, which saves time, improves communication, and makes tracking easier. The technology is available, the timeline is realistic, and the cost is reasonable. Overall, the project is practical, useful, and has a strong potential to improve the internship experience for everyone involved.

Project Plan

ID		Task Mode	Task Name	Duration	Start	Finish	Predecessors	Resource Names
1			**Planning Phase**	**10 days**	**Tue 9/9/25**	**Mon 9/22/25**		
2			information gatheri	7 days				
3			Feasibilty Study	2 days				
4			project plan	1 day				
5			**Analysis Phase**	**18 days**	**Tue 10/7/25**	**Thu 10/30/25**		
6			stakeholder analysi	5 days				
7			gathering requirem	4 days				
8			usecase model	5 days				
9			Data analysis	4 days				
10			**Design Phase**	**15 days**	**Mon 11/3/25**	**Fri 11/21/25**		
11			class diagram mc	6 days				
12			activity diagram r	4 days				
13			system sequence model	4 days				
14			UML diagram	2 days				
15			**Report writing**	**5 days**	**Mon 12/1/25**	**Fri 12/5/25**		
16			documentation	3 days				
17			representation	3 days				
18			**Implementation Phase**	**30 days**	**Sun 12/7/25**	**Thu 1/15/26**		
19			coding	30 days				
20			**Testing Phase**	**10 days**	**Tue 1/20/26**	**Mon 2/2/26**		
21			user testing	5 days				



Project: Project-se
Date: Wed 11/5/25

Task		Inactive Summary		External Tasks	
Split		Manual Task		External Milestone	
Milestone		Duration-only		Deadline	
Summary		Manual Summary Rollup		Progress	
Project Summary		Manual Summary		Manual Progress	
Inactive Task		Start-only			
Inactive Milestone		Finish-only			

Page 1

ID		Task Mode	Task Name	Duration	Start	Finish	Predecessors	Resource Names
22			bug fixes	5 days				
23			**Feedback and Maintanence**	**0 days**	**Tue 2/10/26**	**Tue 2/10/26**		
24			collecting feedback					
25			apply improvments					

1. **Planning Phase (10 days) :** This phase focuses on defining the project's scope and goals. It includes gathering initial information, conducting a feasibility study, and creating a detailed project plan.

   Estimated budget: Project Manager (5,000) , Business Analyst (4,000sar)

2. **Analysis Phase (15 days) :** In this stage, the team gathers and analyzes all functional and non-functional requirements. Stakeholders' needs are translated into use cases and context diagram to ensure clarity on what the system should achieve.

   Estimated budget: Project Manager (9,000sar) , Business Analyst (7,200sar) , System Analyst (7,200sar)

3. **Design Phase (15 days) :** This phase defines the system's architecture and detailed design. UML diagrams (class, activity, sequence) are prepared to represent the structure and interactions of the system components.

   Estimated budget: System Designer/ Architect (7,000sar)

4. **Report Writing (5 days) :** All findings, diagrams, and designs are compiled into a report. This document summarizes the entire design and analysis for presentation and documentation purposes.
   Estimated budget: Project Manager (2,500)

5. **Implementation Phase (30 days) :** Developers start coding the system according to the approved design. The backend, frontend, and database are integrated and tested internally to ensure functionality.
   Estimated budget: Frontend Developer (13,000sar) , Developer Backend (13,000sar) , Database Engineer(12,000sar)

6. **Testing Phase (10 days) :** The developed system goes through testing to identify and fix bugs. User testing is conducted to ensure usability, reliability, and performance before launch.
   Estimated budget: QA Engineer (3,000sar) ,Developer (4,333sar)

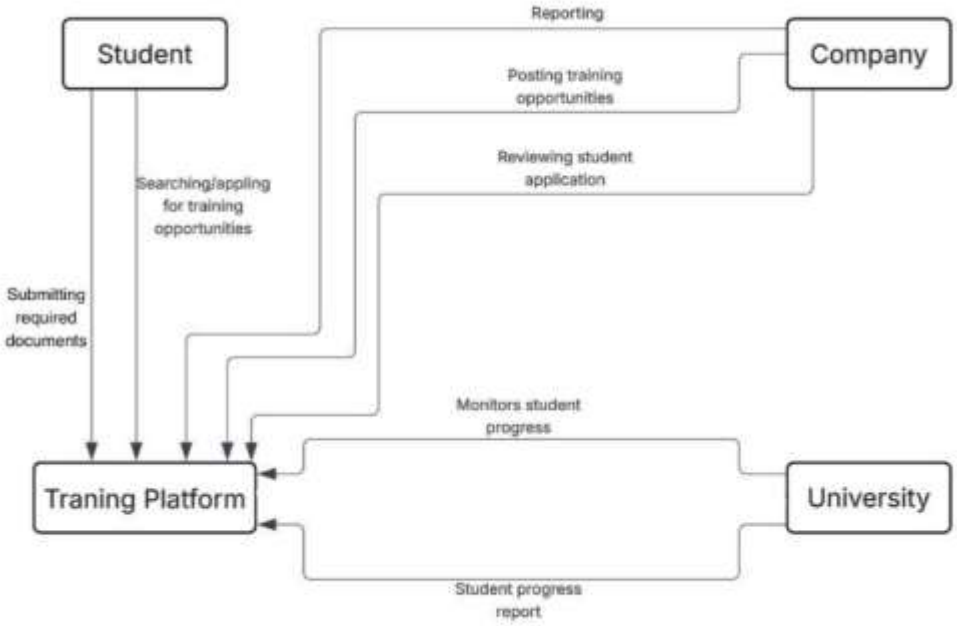
7. **Feedback & Maintenance (5 days) :** After deployment, user feedback is collected and analyzed to make improvements and apply updates. Maintenance ensures system stability and optimization.
   Estimated budget: Developer (2,167sar)

Total Estimated Project Budget : 89,400 SAR

Stakeholders

- Student
- University
- Company

Context Diagram



Event Table

Event name	Input and output	summary
login	Login(in)	Students, university and company can login to the platform
logout	Logout(in)	Students, university and company can logout from the platform
Searching / applying for training opportunities	Searching / applying for training opportunities(in)	Students Searching / applying for training opportunities on the platform
Submitting required document	Submitting CVs(in)	Students submit their CVs on the platform
reporting	Weekly report(in)	Company sending weekly report to the platform
Posting training opportunities	Posting training opportunities(in)	Company Posting available training opportunities on the platform
Reviewing student application	Reviewing student applications (out)	The platform provides a list of students applications to the company
monitoring student progress	monitoring student progress(in)	The university monitoring students progress on the platform
Students progress report	Students progress report(out)	The platform sending each student progress report to the university

Functional Requirements

Students

1. Students can create an account , log in ,and log out.

2. Students can search for training opportunities.

3. Students can submit training applications.

4. Students can track their application status.

5. Students can upload required documents.

6. Students can log training activities.

7. Students can receive evaluations and feedback.

8. Students can download a digital training certificate.

9. Students can view reviews from previous trainees

10.Students can communicate with university staff and company representative.

University Staff

1. University staff can create an account , log in ,and log out.

2. University staff can view all student profiles and statuses.

3. University staff can approve or reject internship applications.

4. University staff can approve or reject company offers.

5. University staff can track student attendance and performance.

6. University staff can generate statistics and reports.

7 . University staff can communicate with students and company representative.

Company Representative

1. Company representative can create account, log in , and log out.
2. Company representative can post training opportunities.
3. Company representative can receive and review student applications.
4. Company representative can accept or reject candidates.
5. Company representative can evaluate interns.
6. Company representative can upload confirmation documents.
7. Company representative can communicate with students and university staff.

System

1. Send Notifications (To cover alerts for applications, evaluations, and communication.)

Non-Functional Requirements

Performance

1. Main pages should load within 3 seconds.
2. System should support at least 500 concurrent users.

Security

1. Users must authenticate with secure login.
2. Sensitive data must be encrypted.

Usability

1. Interface should be simple for all users.
2. System should support Arabic and English.

Availability

System available at least 95% during university hours.

Reliability & Backup

Automatic data backup must be available.

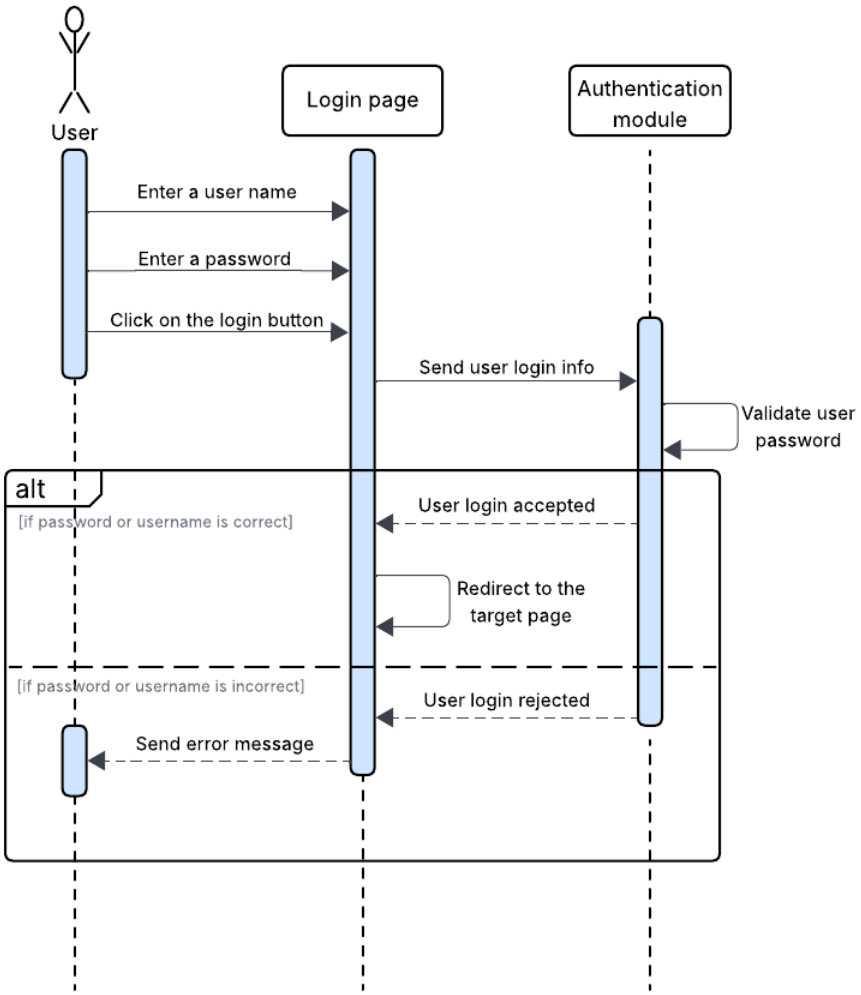
Compatibility

System should work on major browsers and mobile devices

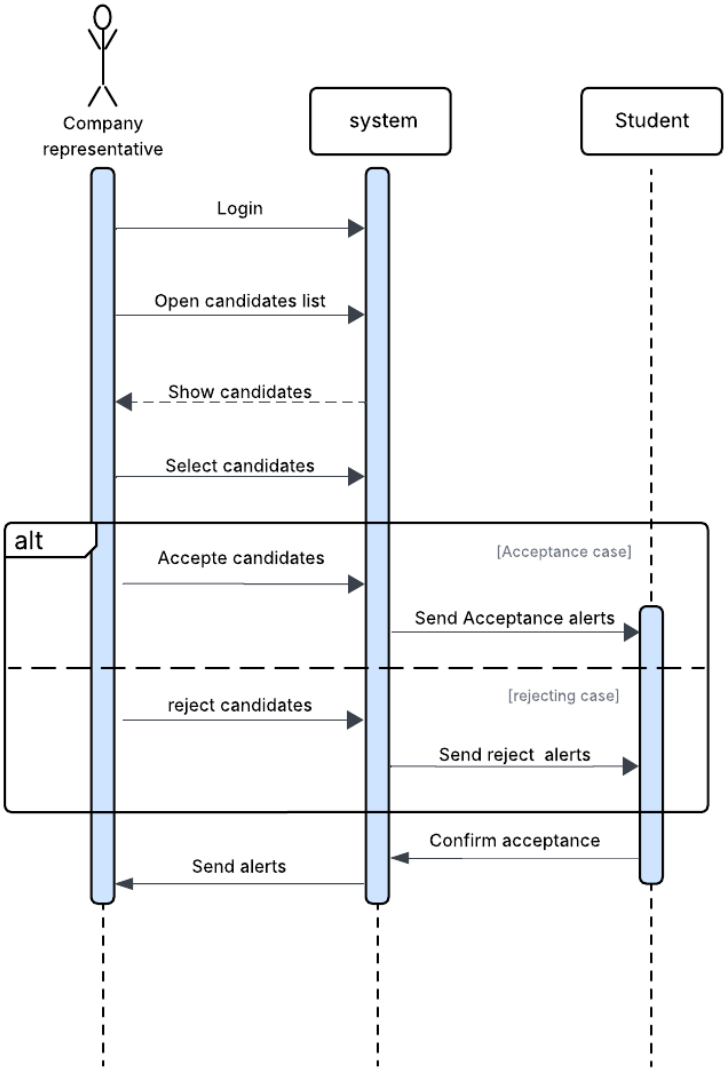
Use Case Diagram

Sequence Diagram

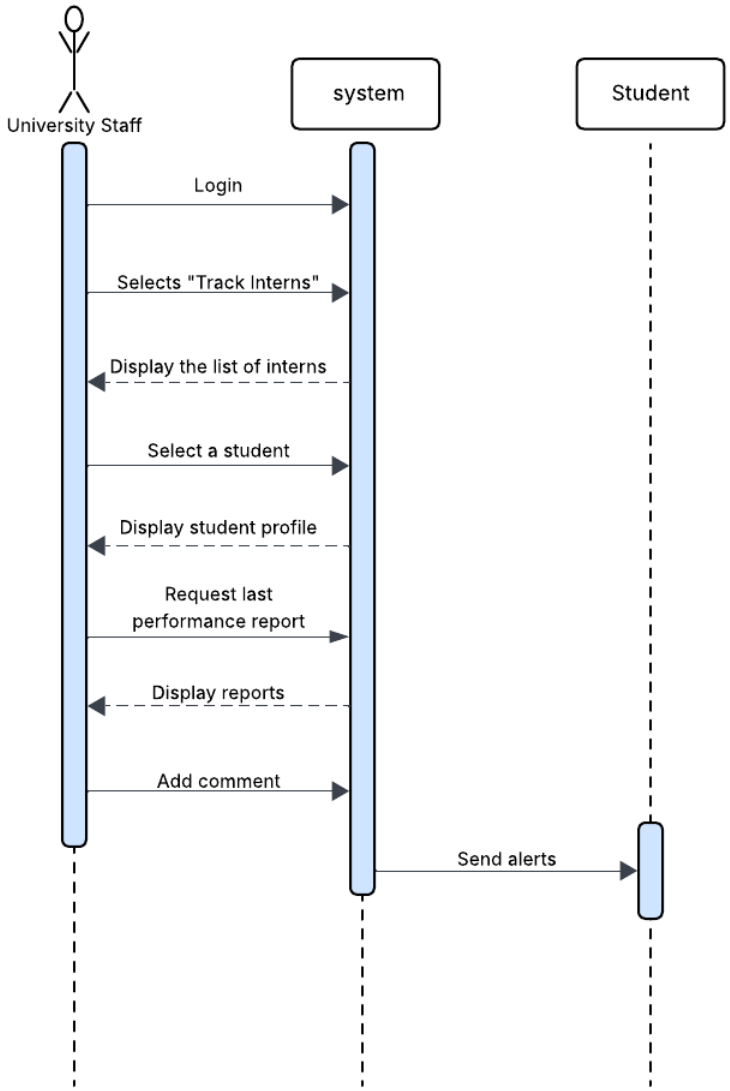
- **Log in**



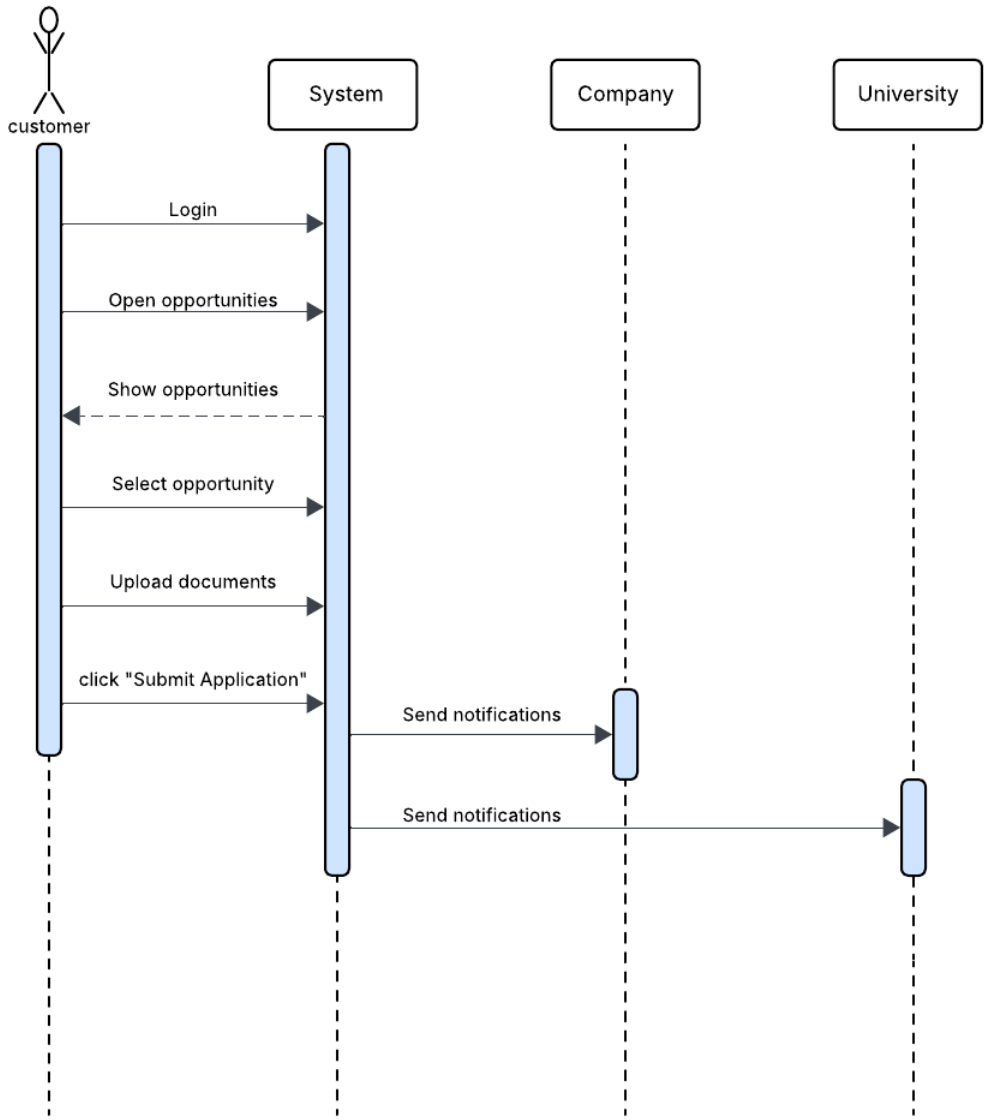
- **Manage Candidates**



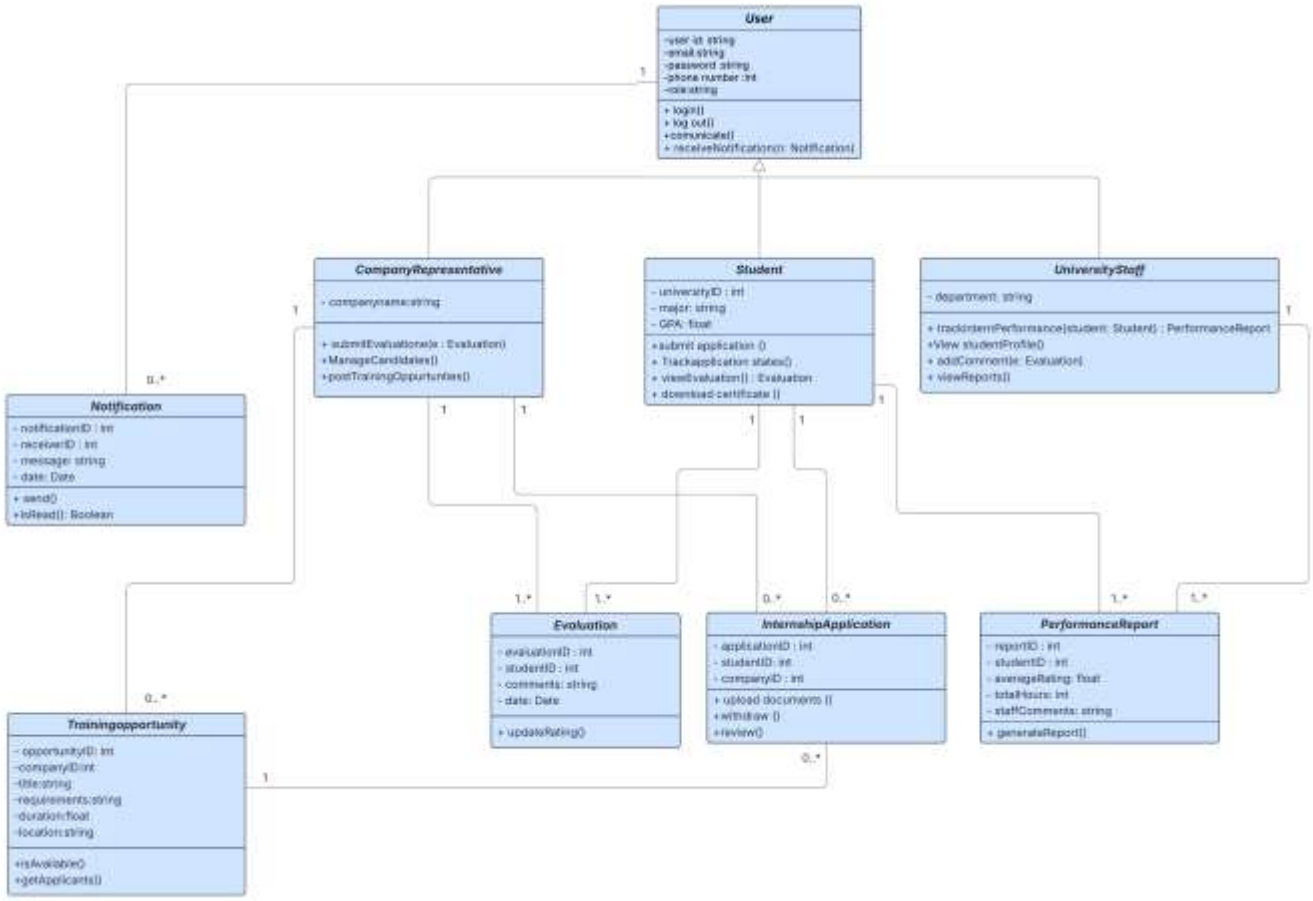
- **Track interns/students**



- **Submit Application**



Class Diagram



1. User

Represents the base class for all system users, including students, university staff, and company representatives it stores common attributes such as user ID, email, password, phone number, and role. The class also handles login/logout functionality, user communication, and receiving system notifications.

2. Student (inherits from User)

Represents a student applying for internship opportunities it stores university ID, major, and GPA. Students can submit internship applications, track their application status, view evaluations submitted by companies, and download their completion certificate after finishing the internship.

3. University Staff (inherits from User)

Represents academic staff responsible for monitoring students during their internships. The class includes the staff member's department and provides functionality to track a student's internship performance, view student profiles, add comments or evaluations, and access performance reports.

4. Company Representative (inherits from User)

Represents a company's HR or recruiter who manages internship opportunities it stores the company name and provides functions to post new training opportunities, manage candidates, and submit performance evaluations after the training is completed.

5. Training Opportunity

Represents a training position offered by a company it includes the opportunity ID, company ID, title, requirements, duration, and location. The class allows checking availability and retrieving all student applicants for the opportunity.

6. Internship Application

Represents a student's application to a specific internship opportunity it stores application details including student ID and company ID. Students can upload required documents, withdraw their application, and review the application status.

7. Evaluation

Represents the company's feedback on a student after the internship it includes evaluation ID, student ID, comments, and date of submission. The evaluation can update the student's rating or performance status within the system.

8. Performance Report

Represents a detailed report generated by the university staff about a student's internship progress. It stores the student ID, average rating, total completed hours, and staff comments. The class can generate a final report summarizing the student's performance throughout the training period.

9. Notification

Represents system generated alerts sent to users for updates it includes notification ID, receiver ID, message content, date, and the read status . The class provides a method to send and mark notifications as read.